

Locating Roots of Equations

Chapter 3.

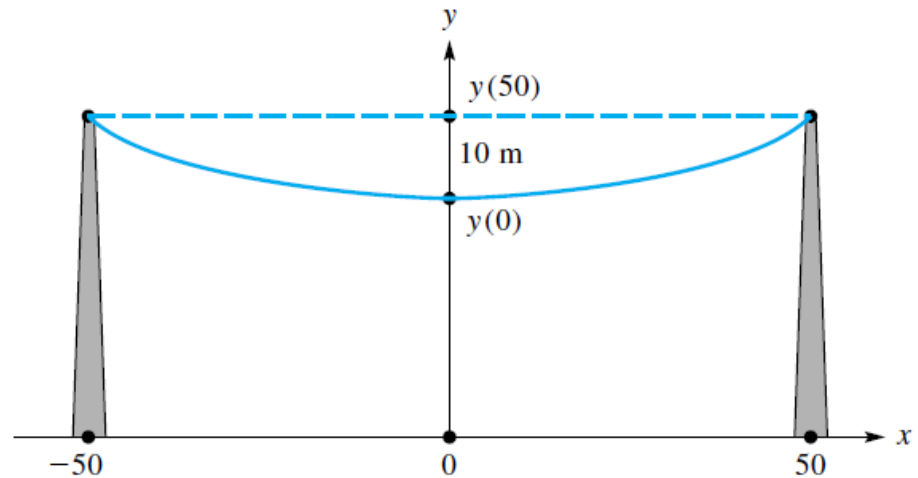
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Introduction

$$f(x) = a \cosh(x/a)$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

hyperbolic cosine



- An electric power cable is suspended from two towers that are 100 meters apart. The cable is allowed to dip 10 meters in the middle. How long is the cable?:
- A hanging cable forms a curve called a **catenary** defined using the **cosh** function: $f(x) = \lambda \cosh(x/\lambda)$. Here λ is a parameter to be determined.
- The conditions of the problem are that $y(50) = y(0) + 10$. Hence, we obtain : $\lambda \cosh(50/\lambda) = \lambda + 10$.
- By the methods of this **chapter**, the parameter is found to be $\lambda = 126.632$. After this value is substituted into the arc length formula of the catenary*, the length is determined to be 102.619 m.

$$* L = 2 \lambda \sinh(50/\lambda)$$

https://www.youtube.com/watch?v=mnBLG_D1nHg

Introduction

- In this chapter we consider one of the most basic problems of numerical approximation, the **root-finding problem**.
- This process involves finding a **root**, or solution, of an equation of the form $f(x) = 0$, for a given function f . A root of this equation is also called a **zero** of the function f .

Chapter's sections

- 3.1 Bisection Method.
- 3.2 Newton's Method
- 3.3 Secant Method
- Summary

Bisection Method



Section 3.1

Introduction

- Let f be a function of a variable. A number r , for which $f(r) = 0$ is called a **root** of that equation or a **zero** of f . For example, the function: $f(x) = 6x^2 - 7x + 2$

- Finding the **zero** of f mean solving the quadratic equation : $6x^2 - 7x + 2 = 0.$ (*)

- The solution to the quadratic equation: $ax^2 + bx + c = 0$ is given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- So the equation has two roots, and depending on the value of the discriminant, $b^2 - 4ac$, the equation may have **real**, **complex** or **repeated** roots.

- If $b^2 - 4ac > 0$ then the roots are real.

- If $b^2 - 4ac < 0$ then the roots are complex.

- If $b^2 - 4ac = 0$ then the roots are real and repeated.

- So, the **zero** of f are $1/2$ and $3/2$. as can be verified by direct substitution or by writing f in its factored form:

$$f(x) = (2x - 1)(3x - 2)$$

Introduction

- For another example, the function

$$g(x) = \cos 3x - \cos 7x$$

- has not only the obvious zero $x = 0$, but every integer multiple of $\pi/5$ and of $\pi/2$ as well, which we discover by applying the trigonometric identity:

$$\cos A - \cos B = 2 \sin \left[\frac{1}{2}(a + b) \right] \sin \left[\frac{1}{2}(b - a) \right]$$

- Consequently, we find

$$g(x) = 2 \sin(5x) \sin(2x)$$

- Why is locating roots important?
- In some problems in which a root of an equation is sought, we can perform the required calculation with a hand calculator. But how can we locate zeros of complicated functions such as these?

$$f(x) = 3.24x^8 - 2.42x^7 + 10.34x^6 + 11.01x^2 + 47.98$$

$$g(x) = 2^{x^2} - 10x + 1$$

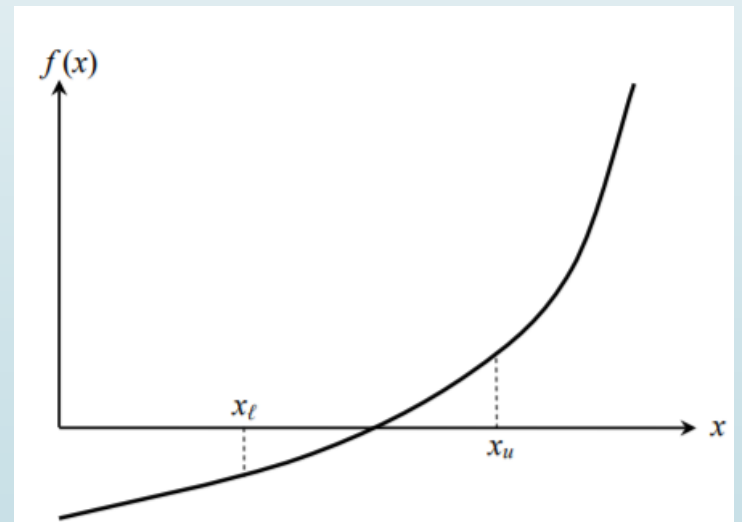
$$h(x) = \cosh \left(\sqrt{x^2 + 1} - e^x \right) + \log |\sin x|$$

Introduction

- What is needed is a general **numerical method** that does not depend on special properties of our functions.
- Hundreds of methods are available for locating zeros of functions, and **three** of the most useful have been selected for study here: the **bisection method**, **Newton's method**, and the **secant method**.
- After reading this section, you should be able to:
 1. *follow the algorithm of the bisection method of solving a nonlinear equation,*
 2. *Use the bisection method to solve examples of finding roots of a nonlinear equation,*
 3. *Follow the algorithm of the False Position (Regula Falsi) Method and Modification.*

Bisection Algorithm and Pseudocode :

- What is the **bisection method** and what is it based on?
- The first technique, based on the *Intermediate Value Theorem*, is called the **Bisection**, or (**Binary-search, method**). It is developed to find the root of a nonlinear equation $f(x) = 0$. The method is *based on* the following theorem:
- **Theorem** : An equation $f(x) = 0$, where $f(x)$ is a real continuous function, has at least one root between x_l and x_u if $f(x_l)f(x_u) < 0$.
- **Note** that if $f(x_l)f(x_u) > 0$, there may or may not be any root between x_l and x_u . So the theorem only guarantees one root between x_l and x_u .
- The **bisection method** exploits this property of continuous functions.



Bisection Algorithm and Pseudocode :

➡ Algorithm of bisection method:

1. At each step in this algorithm, we have an interval $[a, b]$ and the values $u = f(a)$ and $v = f(b)$.

The numbers u and v satisfy $uv < 0$, or in other words, $f(x)$ changes sign between a and b .

2. Next, we construct the midpoint of the interval, $c = \frac{1}{2}(a + b)$, and compute $w = f(c)$.

3. Now check the following:

a) If $uw < 0$, then the root lies between a and c ; then $a = a$ and $b = c$. Go to 2

b) If $uw > 0$, then the root lies between c and b ; then $a = c$ and $b = b$. Go to 2

c) If $uw = 0$; then the root is c . Stop the algorithm if this is true.

Bisection Algorithm and Pseudocode :

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```
procedure Bisection( $f, a, b, nmax, \varepsilon$ )
integer  $n, nmax$ ;  real  $a, b, c, fa, fb, fc, error$ 
 $fa \leftarrow f(a)$ 
 $fb \leftarrow f(b)$ 
if  $\text{sign}(fa) = \text{sign}(fb)$  then
    output  $a, b, fa, fb$ 
    output "function has same signs at  $a$  and  $b$ "
    return
end if
 $error \leftarrow b - a$ 
for  $n = 0$  to  $nmax$  do
     $error \leftarrow error/2$ 
     $c \leftarrow a + error$ 
     $fc \leftarrow f(c)$ 
    output  $n, c, fc, error$ 
    if  $|error| < \varepsilon$  then
        output "convergence"
        return
    end if
    if  $\text{sign}(fa) \neq \text{sign}(fc)$  then
         $b \leftarrow c$ 
         $fb \leftarrow fc$ 
    else
         $a \leftarrow c$ 
         $fa \leftarrow fc$ 
    end if
end for
end procedure Bisection
```

4. Here, the iterations terminate if they exceed $nmax$ or if the error bound (discussed later in this section) is less than ε .

Bisection Algorithm and Pseudocode :

- **Example:** Suppose that we have two functions, and for each, we seek a zero in a specified interval:

```

program Test_Bisection
integer  $n, nmax \leftarrow 20$ 
real  $a, b, \varepsilon \leftarrow \frac{1}{2}10^{-6}$ 
external function  $f, g$ 
 $a \leftarrow 0.0$ 
 $b \leftarrow 1.0$ 
call Bisection( $f, a, b, nmax, \varepsilon$ )
 $a \leftarrow 0.5$ 
 $b \leftarrow 2.0$ 
call Bisection( $g, a, b, nmax, \varepsilon$ )
end program Test_Bisection
  
```

```

real function  $f(x)$ 
real  $x$ 
 $f \leftarrow x^3 - 3x + 1$ 
end function  $f$ 
  
```

```

real function  $g(x)$ 
real  $x$ 
 $g \leftarrow x^3 - 2 \sin x$ 
end function  $g$ 
  
```

$$f(x) = x^3 - 3x + 1 \quad \text{on } [0, 1]$$

$$g(x) = x^3 - 2 \sin x \quad \text{on } [0.5, 2]$$

- In this example, there would be two function procedures and two calls to the bisection procedure.
- The computer results for the iterative steps of the bisection method for $f(x)$:

n	c_n	$f(c_n)$	error
0	0.5	-0.375	0.5
1	0.25	0.266	0.25
2	0.375	-7.23×10^{-2}	0.125
3	0.3125	9.30×10^{-2}	6.25×10^{-2}
4	0.34375	9.37×10^{-3}	3.125×10^{-2}
$\vdots$			
19	0.34729 67	-9.54×10^{-7}	9.54×10^{-7}
20	0.34729 62	3.58×10^{-7}	4.77×10^{-7}

Bisection Algorithm and Pseudocode :

- Also, The results for $g(x)$ as follows:

n	c_n	$g(c_n)$	error
0	1.25	5.52×10^{-2}	0.75
1	0.875	-0.865	0.375
2	1.0625	-0.548	0.188
3	1.15625	-0.285	9.38×10^{-2}
4	1.203125	-0.125	4.69×10^{-2}
$\vdots$			
19	1.2361827	-4.88×10^{-6}	1.43×10^{-6}
20	1.2361834	-2.15×10^{-6}	7.15×10^{-7}

- To verify these results, we use built-in procedures in mathematical software such as **Mat Lab**, or **Mathematica** to find the desired roots of f and g to be 0.34729 63553 and 1.23618 3928, respectively.

Convergence Analysis:

- Now let us investigate the accuracy with which the bisection method determines a root of a function.

➤ Theorem 1:

BISECTION METHOD THEOREM

If the bisection algorithm is applied to a continuous function f on an interval $[a, b]$, where $f(a)f(b) < 0$, then, after n steps, an approximate root will have been computed with error at most $(b - a)/2^{n+1}$.

- If an error tolerance has been prescribed in advance, it is possible to determine the **number of steps** required in the bisection method. Suppose that we want $|r - c_n| < \varepsilon$ then it is necessary to solve the following inequality for n :

$$[b - a] / 2^{n+1} < \varepsilon \quad (r \text{ is a root of } f, c_n = (a_n + b_n)/2)$$

- By taking logarithms, we obtain:

$$n > \frac{\log(b - a) - \log(2\varepsilon)}{\log 2} \quad (2)$$

- **Example :** How many steps of the bisection algorithm are needed to compute a root of f to full machine single precision on a 32-bit word-length computer if $a = 16$ and $b = 17$?

Convergence Analysis:

Solution : The root is between the two binary numbers $a = (10\ 000.0)_2$ and $b = (10\ 001.0)_2$. Thus, we already know five of the binary digits in the answer. Since we can use only 24 bits altogether, that leaves 19 bits to determine.

- We can use Equation (2), which in this case is :

$$n > \frac{\log 1 - \log 2^{-19}}{\log 2}$$

- Then $n \geq 20$.
- In the bisection method, not every guess is closer to the root than the previous guess because the bisection method does not use the nature of the function itself.
- Often the bisection method is used to get close to the root before switching to a faster method.

False Position (Regula Falsi)

Method and Modifications

- In the general case, **false position method** retains the main feature of the bisection method: that a root is trapped in a sequence of intervals of decreasing size.
- Rather than selecting the midpoint of each interval, this method uses the point where the secant lines intersect the x-axis.

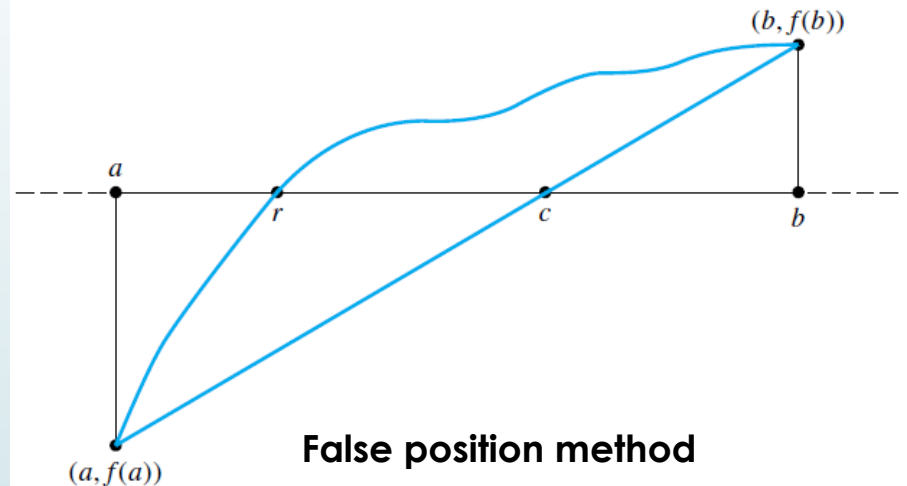
- The two right triangles in the figure are *similar*, which means that :

$$\frac{b - c}{f(b)} = \frac{c - a}{-f(a)}$$

- It is easy to show that:

$$c = b - f(b) \left[\frac{a - b}{f(a) - f(b)} \right] = a - f(a) \left[\frac{b - a}{f(b) - f(a)} \right] = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

- We then compute $f(c)$ and proceed to the next step with the interval $[a, c]$ if $f(a)f(c) < 0$ or to the interval $[c, b]$ if $f(c)f(b) < 0$.



False Position (Regula Falsi) Method and Modifications

➤ **Example :** Lets solve the equation $x^3 - 2x - 3 = 0$, we consider the function $f(x) = x^3 - 2x - 3$ on the interval $[0,2]$.

➤ The function is negative at 0 and positive at 2. this means :

$$f(0).f(2) = (-3).(1) = -3 < 0$$

➤ So, we can apply the algorithm.

$$c = [a.f(b) - b.f(a)] / (f(b) - f(a)) = 3/2$$

➤ Now, $f(c) = f(3/2) = -21/8$

➤ *It is negative and we will make $a = 3/2$ and b is the same, and apply the same thing to the interval $[3/2, 2]$.*

$$c = [(3/2)(1) - (2)(-21/8)] / (1 - (-21/8)) = 54/29$$

➤ Now, $f(c) = f(54/29) = -0.267785$

➤ *It is negative and we will make $a = 54/29$ and b is the same, and apply the same thing to the interval $[54/29, 2]$.*

Summary

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- (1) For finding a zero r of a given continuous function f in an interval $[a, b]$, n steps of the **bisection method** produce a sequence of intervals $[a, b] = [a_0, b_0], [a_1, b_1], [a_2, b_2], \dots, [a_n, b_n]$ each containing the desired root of the function. The midpoints of these intervals $c_0, c_1, c_2, \dots, c_n$ form a sequence of approximations to the root, namely, $c_i = 1/2 (a_i + b_i)$. On each interval $[a_i, b_i]$, the error $e_i = r - c_i$ obeys the inequality

$$|e_i| \leq \frac{1}{2}(b_i - a_i)$$

and after n steps we have

(1).

$$|e_n| \leq \frac{1}{2^{n+1}}(b_0 - a_0)$$

- (2) For an error tolerance ε such that $|e_n| < \varepsilon$, n steps are needed, where n satisfies the inequality

$$n > \frac{\log(b - a) - \log 2\varepsilon}{\log 2}$$

- (3) For the k th step of the **false position method** over the interval $[a_k, b_k]$, let

$$c_k = \frac{a_k f(b_k) - b_k f(a_k)}{f(b_k) - f(a_k)}$$

If $f(a_k) f(c_k) > 0$, set $a_{k+1} = c_k$ and $b_{k+1} = b_k$; otherwise, set $a_{k+1} = a_k$ and $b_{k+1} = c_k$.

